12.1 Counting

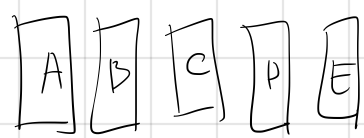
Permutations (order matters)

Combinations (order doesn't matter)

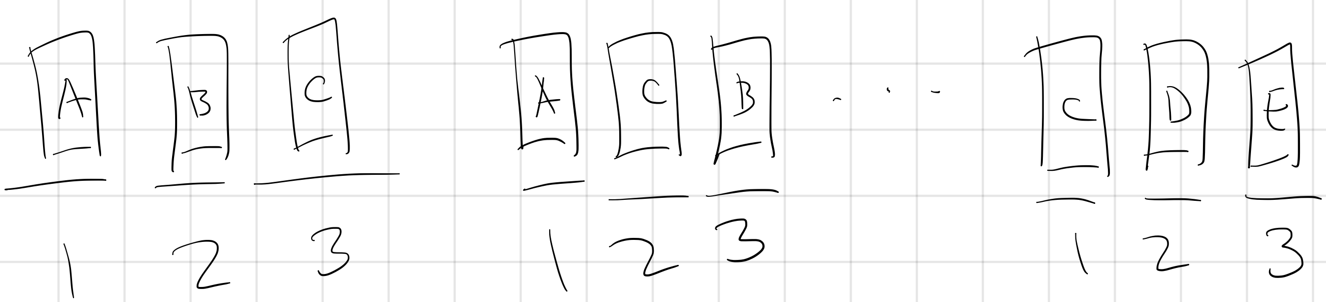
Multiplication

Perm.

5 books



Arrange 3 on a shelf



2 different arrangements

Perm.

5 books

A B C D E

Arrange 3 on a shelf

$$\underbrace{\begin{array}{ccc} A & B & C \\ 1 & 2 & 3 \end{array} \qquad \begin{array}{ccc} A & C & B \\ 1 & 2 & 3 \end{array}}_{\text{2 different arrangements}} \quad \cdots \quad \begin{array}{ccc} C & D & E \\ 1 & 2 & 3 \end{array}$$

Comb.

Take any 3 books

$$\underbrace{A\ B\ C = A\ C\ B}_{\text{same count}}$$

$${}_mP_n \geq {}_mC_n$$

$${}_mP_n = \frac{m!}{(m-n)!}$$

total arrangements for all objects

ways to arrange n objects in m spaces

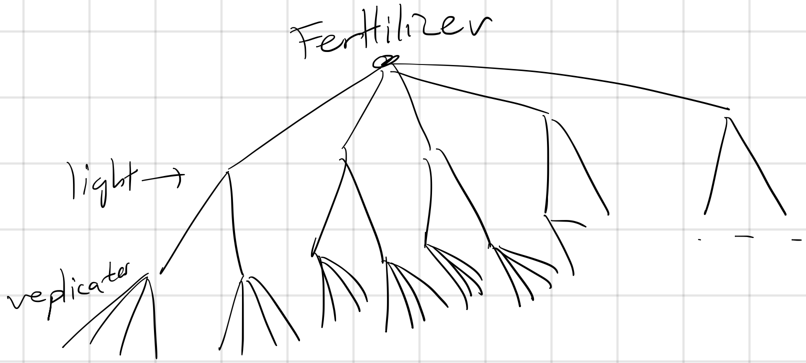
$${}_mC_n = \binom{m}{n} = \frac{m!}{n!(m-n)!}$$

$$_nP_k = \frac{n!}{(n-k)!}$$

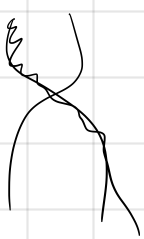
$$_nC_k = \frac{!}{k!(n-k)!} = \binom{n}{k}$$

12.1

1. 5 fertilizers
   2 light levels
   4 experiments each
   $5 \cdot 2 \cdot 4 = 40$



2. 9749 nucleotides
   4 types of nucleotides
   ACTG



$9749 \cdot 4$ different genomes

MISSISSIPPI

M – 1
I – 4
S – 4
P – 2

$$\frac{11!}{\underbrace{4!}_{\text{number of arrangements of I's}}\ 4!\ 2!\ 1!}$$

number of ways of distinct arrangement of letters

$$\begin{array}{ccccccccccc} & 1 & & 2 & & 3 & & & & 4 & \\ \_ & I & \_ & I & \_ & I & \_ & M & \_ & I & \_ \\ 1 & & 2 & & 3 & & 4 & & 5 & & 6 \end{array}$$

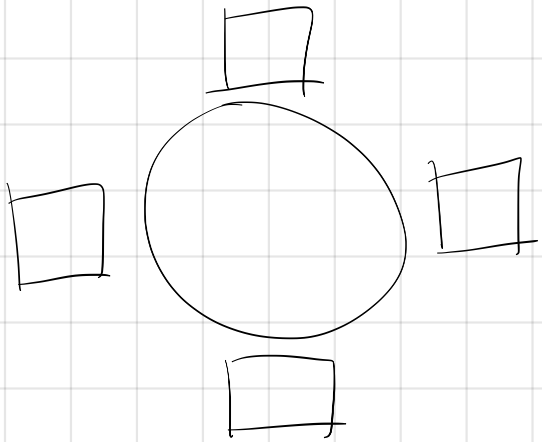
$$\underset{M}{\binom{11}{1}}\underset{I}{\binom{10}{4}}\underset{S}{\binom{6}{4}}\underset{P}{\binom{2}{2}}$$

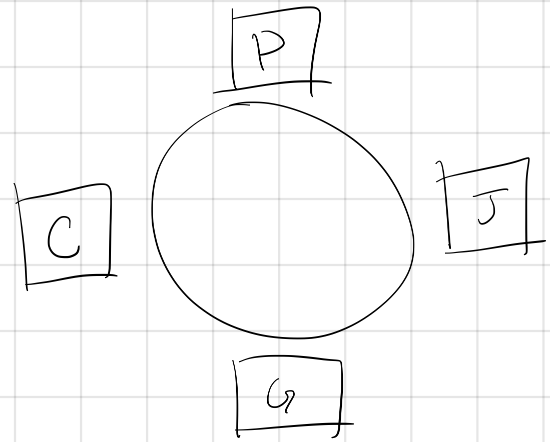
$$\frac{11!}{1!\,\cancel{10!}}\cdot\frac{\cancel{10!}}{4!\,\cancel{6!}}\cdot\frac{\cancel{6!}}{4!\,\cancel{2!}}\cdot\frac{\cancel{2!}}{2!\,0!}$$

$$\frac{11!}{1!\,4!\,4!\,2!}$$

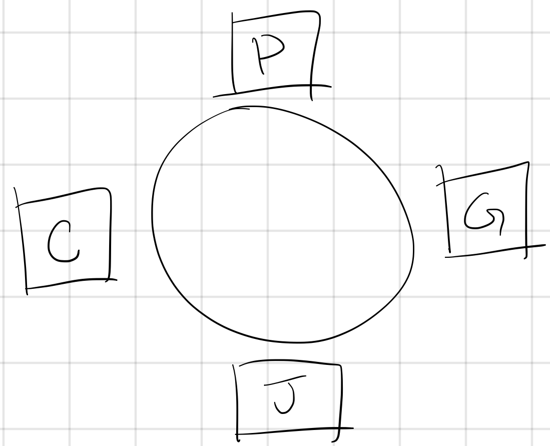
$$\underset{I}{\binom{11}{4}}\underset{M}{\binom{7}{1}}\underset{P}{\binom{6}{2}}\underset{S}{\binom{4}{4}}$$

35. Paula, Cindy, Gloria, Jenny





Cindy left of Paula



4·2 if seat number does matter

2 if seat number does <u>not</u> matter

47. poker

52 cards

4 suits

13 numbers

2 of the same number

$\underline{52} \cdot \underline{3}$

K · K K K

♣ · $\underline{♥}$ $\underline{♠}$ $\underline{♦}$

Binomial Theorem

$$(x+y)^n = \sum_{k=0}^{n} \binom{n}{k} x^k y^{n-k}$$

$(x+y)^0 = 1$ 1

$(x+y)^1 = x+y$ 1 1

$(x+y)^2 = x^2+2xy+y^2$ 1 2 1

1 3 3 1

1 4 6 4 1

Binomial Theorem

44. Expand $(2x-3y)^5$

$1 \quad 4 \quad 6 \quad 4 \quad 1$

$1 \quad 5 \quad 10 \quad 10 \quad 5 \quad 1$

$$(x+y)^n = \sum_{k=0}^{n} \binom{n}{k} x^k y^{n-k}$$

$(x+y)^0 = 1$ $\quad$ 1

$(x+y)^1 = x+y$ $\quad$ 1 1

$(x+y)^2 = x^2+2xy+y^2$ $\quad$ 1 2 1

1 3 3 1

1 4 6 4 1

$$1\cdot(2x)^0(-3y)^5 + 5\cdot(2x)^1(-3y)^4 + 10\cdot(2x)^2\cdot(-3y)^3$$

$$+ 10\cdot(2x)^3\cdot(-3y)^2 + 5\cdot(2x)^4(-3y)^1$$

$$+ (2x)^5(-3y)^0$$

33. Let $S=\{a,b,c\}$ $\quad 2^3=8$

$\{\ \}$ $\qquad \binom{3}{0}$

$+$

$\{a\},\{b\},\{c\}$ $\qquad \binom{3}{1}$

$+$

$\{a,b\},\{b,c\},\{a,c\}$ $\qquad \binom{3}{2}$

$+$

$\{a,b,c\}$ $\qquad \binom{3}{3}$

$$\sum_{k=0}^{3}\binom{3}{k}\cdot 1^{k}\cdot 1^{n-k}=(1+1)^3=2^3=8$$

34. For a set of size n, 2^n